MeetingAI - Code Walkthrough

Every source file explained: what it does, how it works, and how the pieces fit together.

Live application	https://gd4n8x8sjkkxmdhfptkc9m.streamlit.app/
Source code	https://github.com/nidhitiwari23/Meeting-AI-
Size	about 7,700 lines of Python in 50 source files, plus 79 automated tests
Companion document	*MeetingAI - Project Report* (architecture, screenshots, design decisions)

Contents

1 How to read this document

2 The big picture

3 How a Streamlit app runs (and why it matters for the code)

4 Repository layout

5 `app.py` - the entry point

6 `core/` - the application logic (file by file)

7 `ui/` - theme and reusable components

8 `views/` - one file per page

9 Tests

10 Configuration and deployment files

11 Recipes: how to extend the project

12 Design rules and gotchas

---

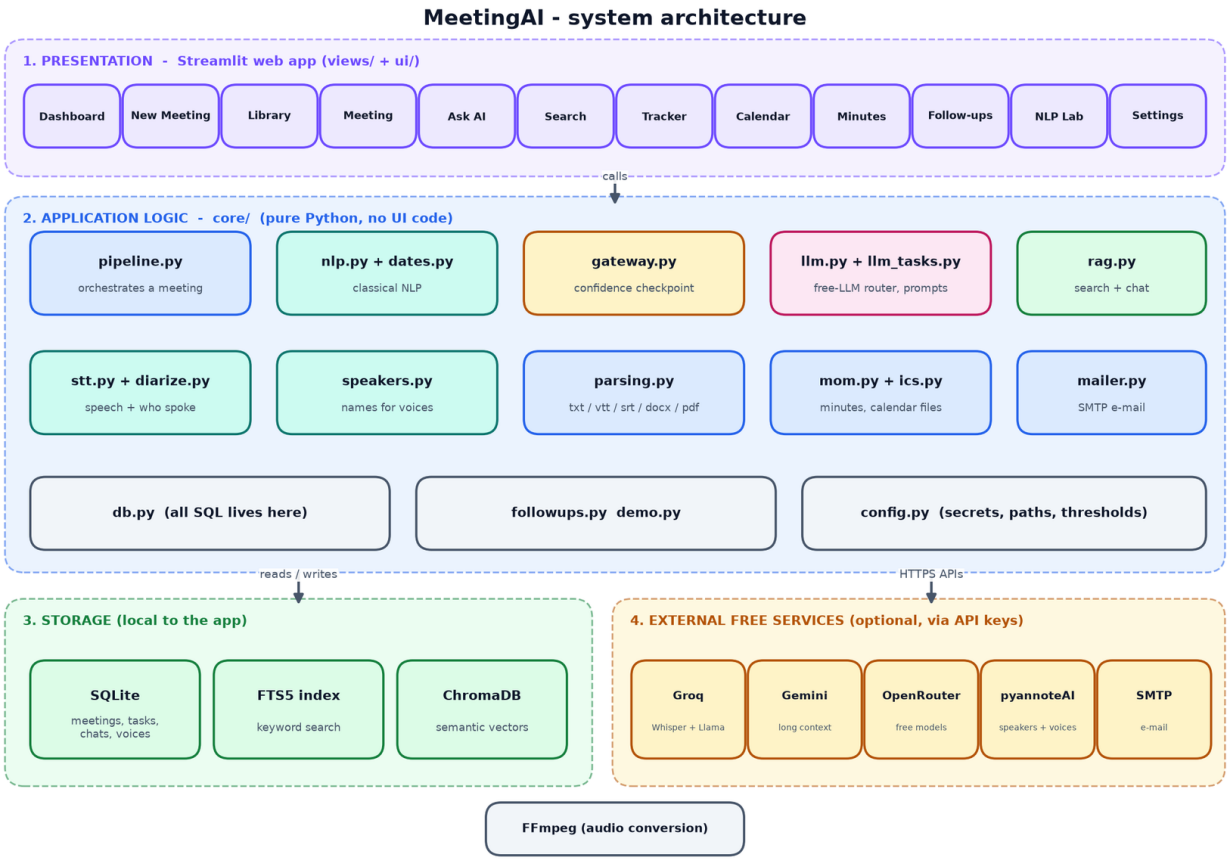
1. How to read this document

Each file has four short parts:

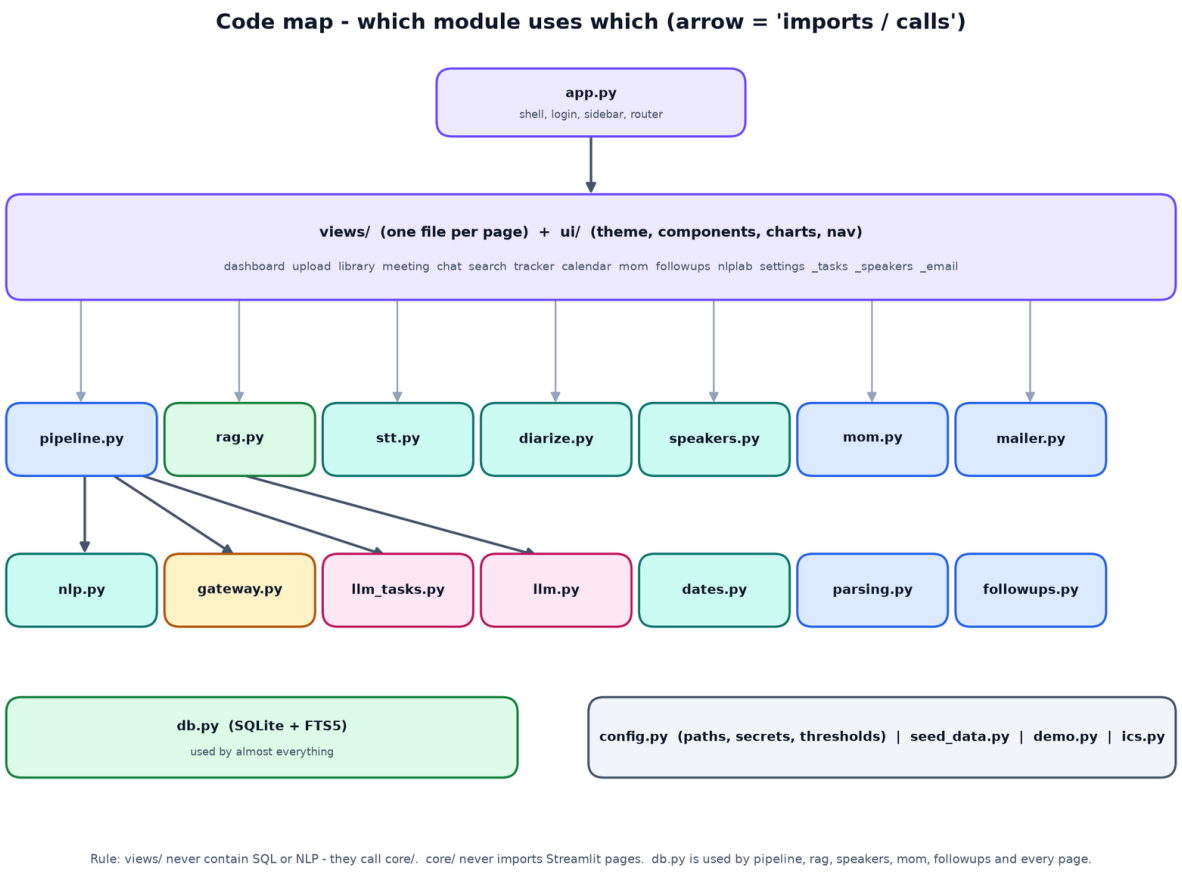
- **Purpose** - what the file is for, in one or two sentences.
- **Key pieces** - the important functions/classes, each with a one-line description.
- **How it works** - the logic in plain language.
- **Talks to** - which other files it uses, so you can follow the flow.

Function names are real; you can search for them in the code.

2. The big picture



System architecture



Which module uses which

Three rules keep the code understandable:

1. `views/` **draws,** `core/` **thinks.** Pages contain no SQL and no NLP; they call functions in `core/`.
2. `db.py` **is the only file that writes SQL** (except small read queries in a few pages).
3. **Everything external is optional.** Each service (LLM, speech, speakers, e-mail) is checked at run time and has a fallback.

3. How a Streamlit app runs (and why it matters)

Streamlit **re-executes the whole script from top to bottom after every click or keystroke**. That has consequences visible all over the code:

- `st.session_state` keeps per-user values between re-runs (current page, selected meeting, chat session, cached suggestions). `ui/nav.py` is just helpers around it.
- **Navigation is a normal function call.** `nav.goto("meeting", meeting_id)` stores the page name and calls `st.rerun()`; `app.py` then imports and renders `views/<page>.py`.
- **Slow, reusable objects are cached** (`@st.cache_resource` for the trained classifier; module-level variables for the embedding model and the ChromaDB client).
- **Durable data goes to SQLite**, never only to session state.
- **Widgets need stable keys** (`key="m_trq"`) so Streamlit does not forget their value when the page re-runs.

- **Long jobs run synchronously** with a progress bar (`views/upload.py`), because the app is a single process on free hosting.

4. Repository layout

```
nlp+agentic ai/
|- app.py                      entry point
|- requirements.txt  packages.txt  requirements-local.txt
|- .streamlit/  config.toml  secrets.toml.example
|- core/    (19 files)  logic: parsing, speech, speakers, NLP, gateway, LLM, search, minutes, e-mail
|- ui/      (4 files)   theme, components, charts, navigation
|- views/   (15 files)  one file per page + 3 shared panels
|- tests/   (9 files)   79 automated tests
`- docs/    guides, reports, diagram and screenshot tools
```

5. `app.py` - the entry point

Purpose. Starts everything: page configuration, theme, optional password, sidebar, and the router that renders the selected page.

Key pieces

Name	Description
`warm()`	`@st.cache_resource`. Creates the database schema and trains the classical NLP model once per server start
`login_gate()`	If the secret `APP_PASSWORD` exists, shows a password form and compares in constant time (`hmac.compare_digest`)
`sidebar()`	Logo, navigation buttons (primary style = current page), status card showing which services are connected, lock button
`main()`	Runs the gate, `warm()`, `sidebar()`, scrolls to the top after navigation (and collapses the sidebar on phones), then imports `views.<page>` and calls `render()`

How it works. `st.set_page_config` must be the first Streamlit call, so it runs before other imports that use Streamlit. `theme.apply()` injects the CSS on every run. The selected page name comes from `ui/nav.current()`. Any exception inside a page is caught and shown as a friendly error plus a technical-details expander, so users never see a raw stack trace.

Talks to. `core.db`, `core.diarize`, `core.llm`, `core.mailer`, `core.stt` (status card), `ui.nav`, `ui.theme`, every `views.*` module.

---

6. `core/` - the application logic

6.1 `core/__init__.py`

A tiny compatibility shim. If the host's SQLite is older than 3.35 (ChromaDB needs newer), it swaps in `pysqlite3` when available. Does nothing on modern systems.

6.2 `core/config.py`

Purpose. One place for paths, defaults and secret lookup.

Name	Description
`DATA_DIR`, `DB_PATH`, `UPLOAD_DIR`, `CHROMA_DIR`, `EXPORT_DIR`	Locations under `data/` (override with the `MEETINGAI_DATA` environment variable - tests use this)
`DEFAULT_THRESHOLDS`	Gateway defaults: classification 0.80, margin 0.15, owner 0.85, deadline 0.85, retrieval similarity 0.28
`DEFAULT_MODELS`	Default free model names (Groq Llama, Gemini Flash, Whisper, OpenRouter)
`secret(name)`	Looks a value up in this order: **session override** (typed in Settings) > **Streamlit secrets** > **environment variable** > default
`set_session_key(name, value)`	Stores a key for the current browser session only

Gotcha handled. Touching `st.secrets` when no secrets file exists makes Streamlit show a red banner, so `_secrets_file_exists()` checks first.

6.3 `core/db.py` - the database layer

Purpose. SQLite is the source of truth. Every table, and every function that reads or writes it, lives here.

Schema (13 tables): `meetings`, `segments` (+ virtual table `segments_fts`), `summaries`, `decisions`, `tasks`, `events`, `mom`, `chat_sessions`, `chat_messages`, `followups`, `analysis_log`, `email_log`, `voiceprints`, `settings`.

Group	Functions
Plumbing	`conn()` (context manager, one short connection per call, thread lock), `init_db()` (creates tables, enables WAL, creates FTS5), `query`, `one`, `execute`
Settings	`get_setting`, `set_setting` (JSON values), `get_thresholds` (defaults merged with saved values)
Meetings	`create_meeting`, `update_meeting` (lists/dicts become JSON), `get_meeting`, `list_meetings` (search/status), `delete_meeting` (cascades to every table and the FTS index)
Transcript	`replace_segments` (also fills FTS), `get_segments`, `rename_speaker` (updates segments, task owners, decisions), `keyword_search`
Insights	`save_summary/get_summary`, `add_decision`, `list_decisions`, `add_task`, `list_tasks`, `update_task`, `add_event`, `list_events`, `clear_auto_items` (removes machine-made items before re-analysis, keeps manual ones)
Documents	`save_mom/get_mom`
Chat	`create_session`, `add_message`, `get_messages`, `list_sessions`, `delete_session`
Follow-ups & logs	`add_followup`, `list_followups`, `log_analysis`, `log_email`
Voices	`save_voiceprint`, `list_voiceprints`, `delete_voiceprint`
Stats	`dashboard_stats()` - all numbers shown on the dashboard

How `keyword_search` works. It turns the query into an FTS5 expression (`"budget" OR "approval"`, or one quoted phrase), runs it against `segments_fts` with `bm25` ranking, and falls back to `LIKE` if FTS5 is unavailable.

6.4 `core/parsing.py` - reading transcripts

Purpose. Turn uploaded text files into `[{speaker, start, end, text}]`.

Name	Description
`parse_transcript(name, bytes)`	Entry point. Detects VTT/SRT by `-->`, otherwise plain text; returns segments + header metadata
`extract_header_meta`	Reads `Meeting:`, `Date:`, `Attendees:`, `Agenda:` lines at the top
`parse_plain`	Recognises `Name: text` with optional `[mm:ss]` timestamps; joins continuation lines; ignores labels like `Note:`/`Agenda:` as speakers; fills `end` from the next start
`parse_cues`	VTT/SRT cues, `<v Name>` voice tags, merges adjacent cues of one speaker
`read_text_file`	Handles `.txt` (several encodings), `.docx` (python-docx), `.pdf` (pypdf)
`to_seconds`, `fmt_ts`, `segments_to_text`	Time helpers and the downloadable transcript format

6.5 `core/stt.py` - speech-to-text

Purpose. Convert audio/video to timestamped text and (optionally) label speakers.

Name	Description
`ffmpeg_path`, `probe_duration`	Locate FFmpeg; read the recording length
`make_chunks`	FFmpeg converts any file to mono 16 kHz 32 kbps MP3 and cuts it into 10-minute pieces so each upload stays under free-tier limits; returns `(file, offset_seconds)`

Name	Description
`_groq_transcribe`	Posts a chunk to Groq's Whisper API asking for segment **and word** timestamps; shifts times by the chunk offset; retries on HTTP 429 using `Retry-After`; drops likely-hallucinated segments
`_assemblyai_transcribe`	Upload, create transcript with speaker labels, poll until done
`_local_transcribe`	faster-whisper or openai-whisper if installed
`merge_short_segments`	Joins Whisper's short fragments into paragraphs (same speaker, small gap, max 60 words)
`choose_provider`, `provider_status`	Decide which engine to use
`transcribe(path, ..., speaker_engine, voiceprints)`	**The orchestrator**: ASR for every chunk, then (if enabled) diarization of the full audio and alignment; returns segments, duration, provider label, identified voices and warnings. If speaker identification fails it keeps a single speaker and returns a warning instead of failing

6.6 `core/diarize.py` - who spoke when (pyannote)

Purpose. Everything about speakers: hosted pyannoteAI, local pyannote.audio, voiceprints, and alignment.

Name	Description
`status()`, `choose_engine()`	Which speaker engines are available; preference order pyannoteAI > local pyannote > AssemblyAI
`_upload`	Declares a `media://` object, receives a pre-signed URL, `PUT`s the audio there
`_run_job`	Submits a job (`diarize`, `identify`, `voiceprint`), polls `/jobs/{id}` until it succeeds or fails; translates HTTP 401/402/429 into clear messages (`DiarizationError`)
`pyannoteai_diarize` / `pyannoteai_identify` / `pyannoteai_voiceprint`	The three hosted operations. *Identify* sends saved voiceprints with a match threshold and returns turns whose speaker is the matched **name** or the generic label
`local_diarize`	Loads `pyannote/speaker-diarization-3.1` with the Hugging Face token (handles old and new argument names) and runs it on a 16 kHz WAV
`diarize()`	Dispatcher used by `stt.transcribe`
`assign_speakers`	**Aligns text with speakers**: each Whisper word goes to the speaker whose turn contains its midpoint; one-word interjections are smoothed away; consecutive words of one speaker become one segment. Without words it uses the largest time overlap
`relabel_generic`	`SPEAKER_00` becomes `Speaker 1`... in order of first appearance; real names are left alone
`extract_voice_sample`	Cuts and joins the longest stretches of one speaker (FFmpeg `atrim` + `concat`) into a clip of at most 28 s - the input pyannoteAI needs for a voiceprint
`trim_clip`	Converts an uploaded/recorded voice clip to a 30-second mono MP3
`match_threshold()`	Strictness (default 60/100), editable in Settings -> Voices

6.7 `core/speakers.py` - putting names on voices

Purpose. Suggest and apply names for anonymous speakers. Nothing is applied without the user's confirmation.

Name	Description
`suggest_from_text`	Regex evidence: **self-introduction** ("I'm Priya", weight 3), **being addressed** ("Rahul, can you..." gives the *next* speaker, 1.2), **being thanked** ("Thanks Rahul" gives the *previous* speaker, 0.9), then **elimination**. Votes are matched greedily one name per speaker; confidence = `0.45 + 0.12 x votes` (max 0.95). Names must be in the attendee list when one exists
`suggest_with_llm`	Optional. Asks the LLM for a mapping **with evidence quotes**, verifies each quote against the transcript and the name against attendees/transcript
`suggest`	Combines both: rules win; agreement raises confidence
`speaker_samples`	The longest lines per speaker, shown beside the naming form
`apply_names`	Renames in segments, task owners and decisions, rebuilds the attendee list and refreshes the vector index

6.8 `core/dates.py` - dates and times

Purpose. Find date expressions and convert them to calendar dates **relative to a base date** (the meeting date). Pure Python, no LLM.

Name	Description
`find_dates(text, base)`	Returns `DateHit(phrase, iso, precision, time, kind)` for every expression: ISO dates, "25th October", "October 25", `13/11`, today/tomorrow/day after tomorrow, "in two weeks", "end of the week/month/quarter", EOD/COB, weekdays ("Monday", "next Friday", "this Friday"), "next week", and vague words
precision	`exact` (a date), `approx` ("next week": text only), `vague` ("soon", ambiguous `03/04`): **no calendar date is invented**
`find_time`	"3 PM", "10:30 am", "noon"
`strip_phrases`	Removes date phrases (and "by/on/before") when building a clean title
`pretty`	`Fri, 16 Oct 2026` or `Not specified`

Weekday rule. "Monday" is the next Monday strictly after the base date; "next Monday" is the Monday of the following calendar week.

6.9 `core/seed_data.py`

Purpose. Training and test sentences for the classifier. `build_train()` expands hand-written templates (`"{P} will {V} {O} {D}."`) with names, verbs, objects and dates using a fixed random seed, giving **610 sentences** in four classes. `TEST_SET` holds **42 hand-written sentences in a different style**, never used for training, for the NLP Lab's evaluation. It is honest about being a *seed* corpus, not real meeting data.

6.10 `core/nlp.py` - classical NLP

Purpose. Everything that understands a sentence without an LLM.

Group	Names	Description
Preprocessing	`clean_for_analysis`, `split_sentences`, `build_sentences`	Remove fillers, split sentences (abbreviation-safe), keep speaker/timestamp/previous speaker/next sentence in a `Sentence` object
Rules	`ACTION_CUES`, `DECISION_CUES`, `HEDGES`, `rule_scores`	Weighted regex cues combined as `1-(1-w1)(1-w2)...`; request wording beats a question mark; hedging detected
Statistics	`Classifier`, `get_classifier`	TF-IDF (1-2-grams) + `MultinomialNB(alpha=0.3)`, plus a logistic regression for comparison; trained from `seed_data` on first use

Group	Names	Description
Fusion	`classify_batch`, `ClassResult`	`0.5 x NB + 0.5 x rules`; hedge x0.8; best label, confidence, margin
Evaluation	`evaluate_classifier`	Accuracy/F1/confusion matrix for NLP Lab
People	`find_people`, `is_real_speaker`, `_match_known`, `extract_owner`	Speaker names (excluding generic "Speaker 1"), first-name matching to full names, patterns for "I'll", "Priya will", "Priya, can you", "X is responsible for", "assigned to X", team names; confidence per pattern
Task details	`make_task_title`, `infer_priority`, `extract_deadline`, `extract_event`	Verb-first titles, High/Medium/Low priority words, deadline selection (prefers a date after "by/before/until", penalises dates from the next sentence), event detection (meeting nouns + exact date + optional time)
Candidates	`candidates_from`, `Candidate`	Applies everything and returns tasks/decisions with `kconf` and `kmargin` for the gateway
Offline summary	`key_terms`, `extractive_summary`	TF-IDF key phrases and centroid-similarity sentence ranking (boosting decisions and actions) used when no LLM is available

6.11 `core/gateway.py` - the confidence gateway

Purpose. Decide whether a classical result is trusted. `evaluate(candidate, thresholds)` returns a `Decision` (accept, reason code, numbers, message). It checks **(1)** `kconf >= cls_threshold`, **(2)** `kmargin >= cls_margin`, then for tasks **(3)** owner confidence `>= owner_threshold`, **(4)** deadline not taken from the neighbouring sentence and exact-date confidence `>= deadline_threshold`. The first failure wins. `REASONS` maps each code to a human sentence. A vague deadline does *not* fail (the LLM cannot invent a date either).

6.12 `core/llm.py` - the free-LLM router

Purpose. Call Groq / Gemini / OpenRouter through their OpenAI-compatible HTTP API using only `requests`.

Name	Description
`_providers()`	Builds the ordered provider list from available keys; Gemini first for long transcripts
`chat()`	Tries each provider and model; skips providers in cool-down; marks a provider as cooling for 45 s after HTTP 429; retries without `response_format` if a model rejects JSON mode; if a model is retired (HTTP 400/404 "model"), calls `discover_model` and tries the replacement once
`discover_model()`	Lists the provider's models and picks a suitable chat model (largest Llama / newest Gemini Flash / `:free` OpenRouter)
`extract_json`, `chat_json()`	Pulls JSON out of fenced or chatty answers, repairs trailing commas, asks the model to fix invalid JSON once
`test_connection()`	Used by Settings -> Test connections
`ACTIVITY`	Small in-memory log of recent calls shown in Settings

6.13 `core/llm_tasks.py` - prompts

Purpose. All prompts and their parsing helpers in one place. `SYSTEM_ANALYST` states the rules (use only the transcript, null when unknown, copy date phrases, quote evidence). `format_transcript` prints `[segment|time] speaker: text`. `chunk_ranges` splits long transcripts by word budget. `analyze_chunk` (summary + items), `merge_summaries` (reduce step), `adjudicate` (batch check of uncertain sentences), `answer` (grounded answers returning JSON with `found`, `answer`, `citations`), `polish_message` (tone-only rewrite).

6.14 `core/pipeline.py` - the orchestrator

Purpose. Run the whole analysis of one meeting and save it.

Name	Description
`analyze(meeting_id, progress, use_llm)`	Steps 1-7 below; returns a `Result` (tasks, decisions, events, log, summary, stats)
`verify_quote`	Fuzzy-matches an LLM quote (RapidFuzz `partial_ratio` >= 82-92) against the segment it claims and then all segments
`validate_owner`	Accepts an owner only if it matches a participant or occurs in the transcript; "I" means the speaker; otherwise *Unassigned* + untrusted
`_resolve_deadline`	Converts an LLM-copied phrase with `dates.py` and checks that the phrase occurs nearby
`_find_match`	Links an LLM item to an existing classical item (same/adjacent segment and similar title)
`_build_summary`	LLM summary (merged for long meetings) or offline extractive summary
`save_results`	Replaces machine-extracted rows, writes the log and summary
`process_meeting`	`analyze` + `save_results` + index in ChromaDB + update attendees/duration/status + refresh follow-ups

`analyze` **in order.** (1) build sentences and candidates; (2) gateway decisions; (3) LLM analysis pass per chunk (summary + independent items); (4) one adjudication call for uncertain candidates; (5) turn candidates into tasks/decisions (accepted, LLM-confirmed, review, or dropped) and log each decision; (6) merge the LLM's independent findings with validation, boosting agreement and flagging conflicts; (7) detect events on *all* sentences (scheduling sentences without a firm commitment go to the calendar, not the task list), de-duplicate tasks, build the summary.

6.15 `core/rag.py` - search and answers

Purpose. The semantic index, hybrid search, query router and grounded answers.

Name	Description
`Embedder`, `get_embedder`	Backends: `onnx` (default, MiniLM, no PyTorch), `sbert` (BGE-small if installed), `hash` (offline lexical fallback). Embeddings are normalised; BGE queries get its instruction prefix
`_collection`	ChromaDB `PersistentClient` collection named per backend, cosine distance
`make_chunks`	About 110 words per chunk with small overlap, remembering segment range, start time and speakers
`index_meeting`, `reindex_items`, `rebuild_all`, `delete_index`	Maintain the vectors (transcript, decision and task documents)
`semantic_search`	Embed the query and query Chroma with optional meeting/kind filters
`route_query`	Detect intents (tasks, decisions, events, summary, phrase), meetings ("last three", a date, a title via fuzzy match) and a person
`hybrid_search`	Semantic + FTS5 keyword, merged with Reciprocal Rank Fusion (`1/(60+rank)`), overlapping ranges merged
`is_relevant`, `lexical_overlap`, `content_tokens`	Relevance gate: similarity threshold or word overlap with light stemming
`answer_question`	Route, retrieve, gate, build labelled evidence ([E], [T], [D], [V], [S]), ask the LLM, verify citations and support, or fall back to a direct retrieval answer

6.16 `core/mom.py` - minutes and document rendering

Purpose. Generate the minutes and render any Markdown to HTML, Word and PDF.

Name	Description
`generate_markdown`	Builds the minutes from saved data (details, attendees, agenda, summary, highlights, decisions table, action-item table, events, conclusions); uncertain rows get a warning mark
`md_blocks`	A small Markdown parser: headings, paragraphs (two trailing spaces = line break), bullet/numbered lists, tables, code blocks, images, page breaks
`to_html`	E-mail/preview HTML with inline styles (links, bold, italic, code)
`to_docx`	python-docx: headings, lists, tables, hyperlinks, images, page breaks
`to_pdf`	ReportLab: styled tables with automatic column widths, images, links; `_pdf_safe` maps symbols the built-in fonts lack
`to_plain`	Plain-text body for e-mail

The same renderer builds the project documents you are reading.

6.17 `core/mailer.py` - e-mail

`parse_recipients` splits free text into valid and invalid addresses; `smtp_config` reads host/port/user/password/security from secrets or the session; `build_message` makes a multipart message (text + HTML + attachments); `send` supports STARTTLS, SSL or plain, never raises and returns `(ok, message)` with a friendly hint for Gmail login errors; `mailto_link` is the no-SMTP fallback.

6.18 `core/ics.py`

`build_ics` writes an iCalendar file (all-day or timed events); `google_calendar_link` builds an "add to Google Calendar" URL so no OAuth is needed.

6.19 `core/followups.py`

`refresh(meeting_id)` regenerates the *suggested* drafts: a recap (decisions + actions), one reminder per owner (a gentler "nudge" when something is overdue), an *owners needed* message for unassigned tasks, and reminders for upcoming events. `polish` asks the LLM to improve tone without changing facts. Drafts are stored as `suggested`; sending happens only in the UI after confirmation.

6.20 `core/demo.py`

Three realistic sample meetings with dates **relative to today**, so the calendar and tracker always have upcoming and overdue items. `load_demo` parses them like uploads and runs the full pipeline.

7. `ui/` - theme and reusable components

File	What it does
`ui/theme.py`	`apply()` injects ~240 lines of CSS: colour variables, navy/violet background gradients, glass cards, sidebar styling, buttons, inputs, tabs, chat bubbles, transcript rows, calendar grid, stepper, animations, and responsive rules (stacked columns and smaller cards on phones)
`ui/components.py`	HTML builders: `H` (collapses indented HTML so Markdown does not treat it as code), `md`, `chip`, `status_chip`, `page_header`, `section`, `metric`/`metric_row`, `empty_state`, `transcript_html` (avatars, timestamps, highlights, insight tags), `month_grid`, `stepper`, `source_card`, `color_for`/`initials` for speaker avatars, `esc` (HTML-escape every piece of user text)
`ui/charts.py`	Plotly helpers (`donut`, `hbar`, `vbar`) with a transparent background, shared palette and fonts
`ui/nav.py`	`PAGES` (sidebar structure), `goto(page, meeting_id, **state)` (set state + `st.rerun`), `current()`

---

8. `views/` - one file per page

Every file exposes `render()`.

File	Page	How it works
`dashboard.py`	Dashboard	`db.dashboard_stats()` drives metric cards; recent meetings, deadlines/events list, task-status donut, owner workload bars, gateway health bars. Empty state offers sample data
`upload.py`	New Meeting	Tabs for file upload and pasted text. `_run()` creates the meeting, parses or transcribes, runs `pipeline.process_meeting` with a live stepper, handles errors by deleting the half-created meeting, and shows a result banner with speaker/AI warnings
`library.py`	Meeting Library	Search, sort, filter; two-column cards; open / ask AI / delete (with confirmation)
`meeting.py`	Meeting details	Tabs: Overview, Transcript (search, speaker filter, insight chips, audio player, download, *Who is who?*), Action items (`_tasks.task_table`), Decisions & events, How it was analysed (gateway log)
`chat.py`	Ask MeetingAI	Session list, scope selector, suggested questions, `rag.answer_question`, typing effect, sources and "how I searched" expanders, saved history
`search.py`	Search	Modes (smart/keyword/semantic/phrase), meeting and date filters, optional grounded answer, highlighted results, related tasks/decisions
`tracker.py`	Action Tracker	Metrics, filters, sorted editable table, manual task form
`calendar.py`	Calendar	Month navigation in `st.session_state["cal_ym"]`, grid from `components.month_grid`, agenda list, Google links, `.ics` download, manual events
`mom.py`	Minutes of Meeting	Generate, Edit/Preview/Export tabs, accept, downloads, recipients validation, e-mail preview, confirmation, send, history
`followups.py`	Follow-ups	Cards for each draft with edit, AI polish, save, mail-app link, dismiss and confirmed send
`nlplab.py`	NLP Lab	Live analysis of pasted text (scores, owner, deadline, event, gateway verdict), evaluation charts and confusion matrix, threshold simulator, gateway statistics, embedding similarity demo
`settings.py`	Settings	AI & speech (keys, test, engines), Voices (`_voices_tab`), Confidence gateway sliders, E-mail, Data & backup (zip backup/restore with path-traversal check, rebuild index, delete all), About
`_tasks.py`	shared	`build_frame` -> `st.data_editor`; `task_table` diffs the edited table against the original and writes only changes (and refreshes the vector index); `add_task_form`

File	Page	How it works
`_speakers.py`	shared	The *Who is who?* panel: suggestions, audio sample players, name inputs, *Remember this voice* (`_enroll` cuts a sample and creates a voiceprint), *Apply names*
`_email.py`	shared	`setup_panel`: inline "connect your e-mail" form with provider presets (Gmail/Outlook/Yahoo)

---

9. Tests (`tests/`)

File	Tests	What it proves
`conftest.py`	-	Isolates every run in a temporary data folder and removes API keys from the environment
`test_dates.py`	8	Weekday and relative-date rules, vague phrases give no date, no false positives
`test_nlp.py`	10	Owners, deadlines, hedging, decisions, events, classifier accuracy floor
`test_parsing.py`	5	Speaker lines, headers, VTT, non-speaker labels, empty file
`test_pipeline.py`	13	Sample meetings end to end, search, "not found", minutes export, **LLM path with a fake LLM** (hallucinated quote and owner rejected)
`test_llm_router.py`	4	Fake HTTP servers: rate-limit fall-through, model discovery, JSON repair, total failure
`test_io.py`	7	Recipient parsing, SMTP against a local server, `.ics`, FFmpeg chunking with a fake ASR
`test_diarize.py`	17	Alignment, fake pyannoteAI server (upload/diarize/identify/voiceprint), failure fallback, voice sample cutting, name suggestions, applying names
`test_ui_smoke.py`	4+	Every page renders (AppTest), chat round trip, *Who is who?* suggests and applies names, Voices tab

Run everything with `python -m pytest tests -q`.

---

10. Configuration and deployment files

File	Purpose
`requirements.txt`	Streamlit, pandas, numpy, plotly, requests, scikit-learn, rapidfuzz, chromadb, onnxruntime, python-docx, reportlab, pypdf
`packages.txt`	`ffmpeg`, installed by Streamlit Cloud
`requirements-local.txt`	Optional `pyannote.audio` + `torch` for local diarization
`.streamlit/config.toml`	Dark theme colours, 200 MB upload limit, no usage statistics
`.streamlit/secrets.toml.example`	Template for all keys (Groq, Gemini, pyannoteAI, AssemblyAI, SMTP, APP_PASSWORD)
`.gitignore`	Keeps secrets, the `data/` folder and caches out of Git
`docs/`	`PROJECT_GUIDE`, `MeetingAI_Project_Report`, this walkthrough (Markdown, Word, PDF), and `tools/` that regenerate screenshots, diagrams and documents

---

11. Recipes: how to extend the project

Add a new page. Create `views/mypage.py` with `render()`; add `("mypage", "My Page", "material_icon")` to `PAGES` in `ui/nav.py`. `app.py` imports it automatically.

Add a new kind of extraction (for example "risks"). Add cue patterns and a label in `nlp.py`, a field/table in `db.py`, a branch in `pipeline.analyze`, and a view. Add seed sentences so the classifier learns the new

label.

Add an LLM provider. Add a dict to `_providers()` in `llm.py` (base URL, key name, models). Anything with an OpenAI-compatible `/chat/completions` endpoint works.

Change the thresholds. Settings -> Confidence gateway (or `DEFAULT_THRESHOLDS`), then use NLP Lab -> Threshold simulator to see the effect.

Use a different embedding model. Set `EMBEDDING_BACKEND` (`onnx`, `sbert`, `hash`); the collection name changes with the backend, then rebuild the index in Settings.

Support another speaker engine. Add a branch in `diarize.diarize()` that returns `DiarizationResult(turns=[{speaker,start,end}], ...)`; alignment and naming work unchanged.

12. Design rules and gotchas

- **Never invent.** Unknown owner = *Unassigned*; unknown date = *Not specified*; no evidence = "not found".
- **Dates are computed by code**, never by an LLM.
- **LLM output is untrusted data** and must pass `pipeline` validation.
- **External actions need a confirmation click** (e-mail, follow-ups, calendar links).
- **Escape user text** with `components.esc` before putting it into HTML.
- **Indented HTML inside Markdown** becomes a code block; `components.H` strips indentation.
- **Hidden control characters.** If a regular expression ever stops matching, check for stray backspace characters (`\x08`) where `\b` was intended.
- **Free hosting forgets files**; the backup/restore feature exists for that reason.
- **Keep views thin.** If a page needs a new rule, put it in `core/` and test it there.